Exercise 1.3

1. The subscription department of a magazine sends out a letter to a large mailing list inviting subscriptions for the magazine. Some of the people receiving this letter already subscribe to the magazine while others do not. From this mailing list, 45% of those who already subscribe will subscribe again while 30% of those who do not now subscribe will subscribe. On the last letter, it was found that 40% of those receiving it ordered a subscription. What percent of those receiving the current letter can be expected to order a subscription?
2. A new transit system has just gone into operation in Chennai. Of those who use the transit system this year, 30% will switch over to using metro train next year and 70% will continue to use the transit system. Of those who use metro train this year, 70% will continue to use metro train next year and 30% will switch over to the transit system. Suppose the population of Chennai city remains constant and that 60% of the commuters use the transit system and 40% of the commuters use metro train next year.
   (i) What percent of commuters will be using the transit system year after the next year?
   (ii) What percent of commuters will be using the transit system in the long run?
3. Two types of soaps A and B are in the market. Their present market shares are 15% for A and 85% for B. Of those who bought A the previous year, 65% continue to buy it again while 35% switch over to B. Of those who bought B the previous year, 55% buy it again and 45% switch over to A. Find their market shares after one year and when is the equilibrium reached?
4. Two products A and B currently share the market with shares 50% and 50% each respectively. Each week some brand switching takes place. Of those who bought A the previous week, 60% buy it again whereas 40% switch over to B. Of those who bought B the previous week, 80% buy it again where as 20% switch over to A. Find their shares after one week and after two weeks. If the price war continues, when is the equilibrium reached?

Exercise 1.4

Choose the correct answer

1. If $A=(1\ \ 2\ \ 3)$, then the rank of AA^T is
   (a) 0 (b) 2
   (c) 3 (d) 1
2. The rank of $m\times n$ matrix whose elements are unity is
   (a) 0 (b) 1
   (c) m (d) n
3. If $T=\begin{matrix} \\ A \\ B \end{matrix}\begin{matrix} \begin{matrix} A & B \end{matrix} \\ \begin{pmatrix} 0.4 & 0.6 \\ 0.2 & 0.8 \end{pmatrix} \end{matrix}$ is a transition probability matrix, then at equilibrium A is equal to
   (a) $\frac{1}{4}$ (b) $\frac{1}{5}$
   (c) $\frac{1}{6}$ (d) $\frac{1}{8}$
4. If $A=\begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix}$, then $\rho(A)$ is
   (a) 0 (b) 1
   (c) 2 (d) n
5. The rank of the matrix $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 4 & 9 \end{pmatrix}$ is
   (a) 0 (b) 1
   (c) 2 (d)3
6. The rank of the unit matrix of order n is
   (a) $n-1$ (b) n
   (c) $n+1$ (d) n^2

XII Std - Business Maths & Stat EM Chapter 1.indd 20 2/28/2020 5:28:20 PM